MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026
Semester V
SEC – MG5SECZGY300,MG5SECZIM300
FOOD AND WATER QUALITY MANAGEMENT
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which toxic metal is associated with Minamata disease? [U] / [CO1]

   a) Lead b) Mercury c) Cadmium d) Arsenic

2. Which of the following is a natural preservative? [U] / [CO1]

   a) Vinegar b) Tartrazine c) Saccharine d) BHA

3. The Food Safety and Standards Act enacted in [K] / [CO2]

   a) 1954 b) 1986 c) 2006 d) 2011

4. Which body regulates food safety in India? [K] / [CO2]

   a) BIS b) AGMARK c) FSSAI d) NABL

5. Agmark certification is mainly used for. [U] / [CO2]

   a) Electronic Goods b) Agricultural Products c)Medicines d) Textiles

6. Which among the following is an intrinsic factor of food spoilage [U] / [CO3]

   a) Temperature b) Relative humidity c) pH d) Presence of bacteria

7. Bacteria causing spoilage of vegetables. [U] / [CO3]

a) *Erwinia* b) *Corynbacterium* c) *Yersinia* d) *Shigella*

8. Which among the following water quality parameter is measured by gravimetric method? [U] / [CO3]

a) Turbidity b)Temperature c)Total Dissolved Solids d) Total hardness

9. Which spoilage causes flat cans, black discoloration of food, and a rotten egg smell? [U] / [CO4]

a) Hydrogen Swell b)Putrefactive anaerobes c)T.A. Spoilage
d) Sulphide stinker spoilage

10. The principle behind freeze drying is. [U] / [CO 4]

a) Coagulation b) Sublimation c) Autolysis d) Desorption

11. The Principle of drying relies on the simultaneous transfer of which two factors? [U] / [CO4]

a) Heat and Mass b) Temperature and Pressure c) Velocity and Volume
d) Moisture and Density

12. FDA approved material widely used for shrink packaging in the food industry is. [U] / [CO4]

a) POF b) PLA c) PHA d) PVC

[1x12 = 12]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. What are the acute health effects of food adulteration? [U] / [CO1]

14. What are artificial sweeteners? Give two examples and mention their uses. [U] / [CO1]

15. Explain the objectives of FSSA 2006. [U] / [CO2]

16.	Explain the scope and significance of quality management system.	[U / [CO2]
17.	What is physical spoilage of food? Give examples.	[U] / [CO3]
18.	Explain microbial spoilage of Dairy products.	[U] / [CO3]
19.	What is Cryogenic freezing? Mention its principle and advantages.	[U] / [CO4]
20.	Distinguish between intelligent packaging and active packaging	[U] / [CO4]

[3x6 = 18]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

21.	Discuss the common food adulterants and methods used for their detection.	[U] / [CO1]
22.	Describe the major provisions of FSS Regulations 2011.	[U] / [CO2]
23.	Explain the intrinsic factors of food spoilage	[U] / [CO3]
24.	Explain the Physical parameters of water quality analysis	[U] / [CO3]
25.	Explain the factors affecting the rate of Drying and Dehydration	[U] / [CO4]
26.	Differentiate between Modified Atmosphere Packaging (MAP) and Controlled Atmosphere Packaging(CAP)	[U] / [CO4]

[5x4 = 20]